FINAL VALUE THEOREM AND STEADY-STATE ERROR

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1.0 Learning Outcomes

Upon completing this practical exercise, students should be able to demonstrate the Learning Outcomes presented in Table 1. At the end of the exercise, revisit each Learning Outcome and confirm that it has been achieved.

Table 1: *Learning Outcomes for the Practical Exercise*

ID	Learning Outcome	Achieved
1	Apply the Final Value Theorem to predict the steady-state error of a closed-loop speed controller, and check the conditions for its use.	☐
2	Measure the steady-state error of the DC motor under proportional control for several gains and compare it with the prediction.	☐
3	Show by prediction and measurement that integral action removes the steady-state error to a step, and relate this to the system type.	☐

2.0 Hardware Set-Up

This practical exercise builds on the hardware and algorithm set-up described in the practical exercise, DC Motor Speed Measurement using an Encoder, which details the operation of a DC motor with an encoder and the measurement of motor speed in revolutions per minute (RPM). The set-up is not repeated here.

The first-order model of the motor, $G(s) = K/(\tau s + 1)$, identified in the practical exercise, Parameter Estimation of a DC Motor using Linear Least Squares, is used throughout. Use the values of K and τ identified for your own motor. Where example values are needed, this guide uses the A1.4 example results, $K = 28$ RPM/V and $\tau = 0.15$ s.

3.0 Practical Investigation via Simulink

MATLAB and Simulink should be used to complete this practical exercise. The Final Value Theorem is used to predict the steady-state error of a speed control loop before the error is measured, first with proportional control and then with PI control. The exercise builds on the practical exercise, Block Diagram Algebra:

Closed-Loop P Speed Control, and supports the ACE-CORE learning guide, Final Value Theorem.

Predict the Steady-State Error

For unity feedback with controller $C(s)$ and motor $G(s)$, the error is $E(s) = R(s)/(1 + C(s)G(s))$. The Final Value Theorem gives its steady-state value directly from the s-domain, provided every pole of $sE(s)$ lies in the left half-plane:

$$e_{ss} = \lim_{t\to\infty} e(t) = \lim_{s\to 0} s\,E(s)$$

For proportional control, $C(s) = K_p$, and a step setpoint of size R:

$$e_{ss} = \lim_{s\to 0} s \cdot \frac{1}{1 + K_p \dfrac{K}{\tau s + 1}} \cdot \frac{R}{s} = \frac{R}{1 + K_p K}$$

For PI control, $C(s) = K_p + K_i/s$. The integrator makes $C(s)G(s)$ infinite at $s = 0$, so $e_{ss} = 0$. The loop has become type 1: it has one integrator, and tracks a step with zero error.

1. Derive the error transfer function and the two results above by hand.
2. Predict the steady-state error for a 50 RPM setpoint with K_pK = 0.5, 1, 2 and 4, and enter the values in Table 2. For the example motor, K_p = 0.018, 0.036, 0.071 and 0.143 V/RPM.
3. Apply the FVT to a ramp setpoint, $R(s) = R/s^2$, with P and with PI control. What do you predict?

Measure the Steady-State Error

1. Use the proportional control model from the practical exercise, Block Diagram Algebra: Closed-Loop P Speed Control. Apply a 50 RPM setpoint and measure the final speed, averaging over the last second, for each K_p.
2. Replace the gain with a PI controller, using K_p = 0.036 V/RPM and K_i = 0.24 V/(RPM·s), and repeat the test.
3. Record the results in Table 2. At the highest gain the controller output may saturate briefly at the start of the step. This does not affect the steady-state value.

Example results are shown in Figure 1. They are simulated with the motor model identified in the practical exercise, Parameter Estimation of a DC Motor using Linear Least Squares (K = 28 RPM/V, τ = 0.15 s), so they show the behaviour to expect rather than measured data. The final speed under P control matches $50K_pK/(1 + K_pK)$ exactly in the simulation, and the PI controller reaches the setpoint.

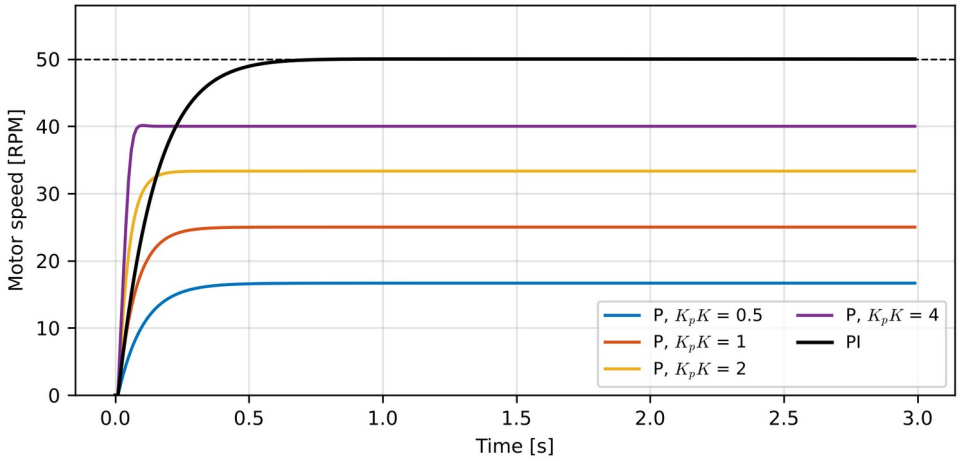


Figure 1: *Simulated Example of the Response to a 50 RPM Setpoint under P Control at Four Gains and under PI Control*

Table 2: *Predicted and Measured Steady-State Error for a 50 RPM Setpoint*

Controller, K_pK	K_p (V/RPM)	Predicted e_{ss} (RPM)	Predicted e_{ss} (%)	Measured final speed (RPM)	Measured e_{ss} (RPM)
P, 0.5	0.018	33.3	67		
P, 1	0.036	25.0	50		
P, 2	0.071	16.7	33		
P, 4	0.143	10.0	20		
PI	0.036	0	0		

The predicted values use the example motor. Recalculate them with your own K. The measured error usually follows the $1/(1 + K_pK)$ trend, but is somewhat larger at low gains, where the motor runs slowly and friction and the dead-zone matter most.

1. What is the type of the open loop with P control, and with PI control?
2. Give an example of a closed-loop error $E(s)$ for which the FVT would give a wrong answer, and explain why.
3. Why does the measured error stop following the prediction at very high K_p?

4.0 Concluding Remarks

This practical exercise used the Final Value Theorem to predict the steady-state error of a DC motor speed controller and tested the predictions on hardware. Using MATLAB, Simulink and ACE-Lab hardware, the error under proportional control was shown to fall as $1/(1 + K_pK)$, and integral action was shown to remove it.

The FVT gives the steady-state behaviour of a stable loop without solving for the full time response. It also explains why integral action is so widely used: adding an integrator raises the system type and removes the steady-state error to a step, whatever the exact gain of the process.

Recommended Citation

If you have used this practical exercise in your teaching, learning or research, please cite it using the reference below, adapting the format as required by your preferred referencing style.

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